and solve the latter exactly. It is not technically necessary to require symmetry of the way that stochastic and deterministic terms are interleaved if the only purpose is equilibrium sampling, although symmetry is required within any splitting method used for the deterministic part in order to gain even order (in our case an $\mathcal{O}(h^2)$ error) in short time approximation and in the accuracy of computed thermodynamic quantities.

In the case of (8.75)–(8.79) this suggests

$$\begin{aligned}
\xi &:= \xi + (\Delta t/2)h_{\mathrm{T}}, \qquad \eta := \eta + (\Delta t/2)h_{\mathrm{P}},\\
\boldsymbol{p} &:= e^{-(\xi+\eta)\Delta t/2}\boldsymbol{p} + \psi(-\xi-\eta, \Delta t/2)\boldsymbol{F}(\boldsymbol{q}),\\
\boldsymbol{q} &:= e^{\eta\Delta t}\boldsymbol{q} + \psi(\eta, \Delta t)\boldsymbol{M}^{-1}\boldsymbol{p},\\
\boldsymbol{p} &:= e^{-(\xi+\eta)\Delta t/2}\boldsymbol{p} + \psi(-\xi-\eta, \Delta t/2)\boldsymbol{F}(\boldsymbol{q}),\\
V &:= e^{\eta\Delta t}V,\\
h_{\mathrm{T}} &:= h_{\mathrm{T}}(\boldsymbol{p}), \qquad h_{\mathrm{P}} := h_{\mathrm{P}}(\boldsymbol{q}, \boldsymbol{p}, V),\\
\xi &:= \xi + (\Delta t/2)h_{\mathrm{T}}, \qquad \eta := \eta + (\Delta t/2)h_{\mathrm{P}},\\
\xi &:= e^{-\gamma_{\mathrm{T}}\Delta t}\xi + \sqrt{\beta^{-1}(1-e^{-2\gamma_{\mathrm{T}}\Delta t})/\mu_{\mathrm{T}}}R_\xi,\\
\eta &:= e^{-\gamma_{\mathrm{P}}\Delta t}\eta + \sqrt{\beta^{-1}(1-e^{-2\gamma_{\mathrm{P}}\Delta t})/\mu_{\mathrm{P}}}R_\eta,
\end{aligned}$$

where

$$\psi(\alpha, t) \stackrel{\text{def}}{=} \begin{cases} \frac{e^{\alpha t}-1}{\alpha} & \alpha \neq 0 \\ t & \alpha = 0 \end{cases},$$

and R_ξ, R_η are two scalar i.i.d. $\mathcal{N}(0,1)$ random numbers. Periodic boundary conditions are enforced whenever $\boldsymbol{q}$ or V is updated, to ensure that the system remains inside the cube with side length $V^{1/3}$.

We perform an experiment with the given discretization for 64 atoms in a 3D periodic box using a pairwise Lennard-Jones potential. We set $\mathrm{P} = 0.5$, $\mathrm{k_B T} = 0.1$ and $\mu_{\mathrm{P}} = \mu_{\mathrm{T}} = 5$, with all particle masses set to one and $h = 0.001$. As plotted in Fig. 8.6, we initialize the particles in a cubic formation within a $5 \times 5 \times 5$ periodic box (so $V(0) = 125$). As we run the simulation, V decreases and the pressure is increased.

We set $\gamma_{\mathrm{T}} = \gamma_{\mathrm{P}} = \gamma$ and compare simulations using three values of γ. Choosing $\gamma = 0$ removes any stochastic terms from the simulation, reducing the dynamics to a similar form to the Nosé-Hoover thermostat. The distribution of the pressure Π, plotted in Fig. 8.7 (centered on the target pressure P), is well sampled under all three choices of the friction γ. However, as in the NVT experiment in Fig. 8.2, without any stochastic terms in the equations of motion the system exhibits a pronounced